

how to develop llm with feel will(1)



momo

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1. Why This Discussion Exists

Current LLM research is extremely successful, yet many fundamental questions remain unresolved. We know how to scale models. We know how to collect data. We know how to optimize objectives. What we do not know is whether these methods are sufficient for understanding intelligence itself.

Most frontier work today follows a bottom-up paradigm: larger datasets, larger models, larger compute budgets, and better optimization procedures. The central question of this seminar is whether this paradigm is enough, or whether some aspects of intelligence require a top-down perspective rooted in philosophy, art, cognition, anthropology, and the history of civilization.

2. Verification Is Not the Same as Existence

One question I find interesting is the relationship between verification and existence.

Suppose consciousness cannot currently be verified.

Suppose free will cannot currently be measured.

Does that imply they do not exist?

Deep learning provides an interesting analogy. Neural networks operate in highly non-convex optimization landscapes, yet they optimize successfully. Practical success arrived long before complete theoretical understanding. Theory followed practice.

This raises the possibility that some phenomena may emerge before we know how to verify them.

3. Hallucination as an Information Problem

My current intuition is that hallucination may partly arise because a model is forced to generate an answer even when the information available to it is insufficient.

This leads to a speculative question:

What happens if the information quality of the input is comparable to the information quality expected in the output?

Would hallucination decrease?

Under this view, hallucination becomes less about factual mistakes and more about a mismatch between information supply and information demand.

4. Where Does Civilization Store Its Most Valuable Information?



My personal view is that the traditional hierarchy of importance may be backwards.

A tentative ordering might be:

Art → Literature → Social Science → Natural Science → Researchers → Engineers.

The point is not that engineers are less important.

The point is that engineering often implements ideas that originate elsewhere.

Art, literature, philosophy, and culture may contain highly compressed forms of human experience accumulated over centuries.

The real question is:

Where does civilization store its highest-density information?

5. Art as Compressed Knowledge

In machine learning we often observe that useful information is compressed into a surprisingly small portion of a representation.

Perhaps civilization works similarly.

Different people can interpret the same artwork differently, yet meaningful structure remains.

Art may therefore be viewed as a highly compressed representation of human understanding.

If intelligence itself is compressed somewhere inside civilization, then perhaps scaling datasets indefinitely is not the only path forward.

Perhaps the more important task is identifying the small amount of information that matters most.

6. Philosophy as a Navigation System

When I refer to philosophy, I do not mean abstract discussion disconnected from reality.

I mean something closer to guiding principles.

The analogy I like comes from Zhuangzi's story of Cook Ding.

Mastery comes not from brute force but from understanding structure.

Engineering is implementation.

Philosophy is navigation.

Both are necessary, but navigation determines direction.

Current AI evaluates progress using benchmarks.

- AIME
- coding tasks
- RLHF objectives
- reward models
- preference datasets

These metrics are useful.

However, they are not obviously equivalent to intelligence.

If consciousness, agency, or free will cannot currently be measured, why are we so confident that benchmark optimization is the correct route toward them?

This remains an open question.

8. User Experience as an Alternative Signal

One possibility is that end-user experience contains information missing from benchmarks.

Suppose millions of users consistently report:

- this model reasons better,
- this model understands better,
- this model feels more intelligent.

Perhaps such observations deserve serious consideration rather than being dismissed as subjective.

In some sense, this resembles RLHF operating at the scale of civilization.

9. Chaos and the Approximation of Truth

A perspective I find interesting comes from discussions about chaos and society.

Human civilization appears chaotic.

Artists, scientists, entrepreneurs, engineers, and researchers all move in different directions.

Most attempts fail.

Some succeed.

Yet collectively civilization advances.

Academia may work similarly.

Every discipline may contribute part of the answer.

Truth is not necessarily approached through a single path.

The paper *Understanding Deep Learning requires Rethinking Generalization* remains one of my favorite examples of academic taste because it asks a very simple question:

Deep learning works. Why?

Many important research questions begin in exactly this way.

Likewise, controversial works such as *Opening the Black Box of Deep Neural Networks via Information* remind us that important questions often arrive before satisfactory answers.

Right now it jumps:

Truth → Art → Philosophy → Intelligence → LLM

too quickly.

The missing idea is:

Why should LLM researchers care about civilization at all?

Without that bridge, people will think this is random philosophy.

The bridge is actually hidden throughout your notes:

Civilization as Humanity's Compression Algorithm

The reason I keep returning to art, philosophy, literature, science, and social science is not because I believe these disciplines are superior to engineering. The reason is that civilization itself can be viewed as a giant information-compression process.

Every generation produces an enormous amount of experience. Most of that experience disappears. A tiny fraction survives. The surviving fraction becomes stories, literature, philosophy, scientific theories, cultural norms, mathematical abstractions, and eventually engineering practices.

In that sense, civilization is continuously performing representation learning.

What survives is not necessarily what is true, but what is useful enough to be remembered and transmitted.

Art survives because it compresses human experience.

Philosophy survives because it compresses patterns of thought.

Science survives because it compresses regularities in nature.

Mathematics survives because it compresses logical structure.

Engineering survives because it compresses successful implementation.

This observation is interesting because modern machine learning appears to operate in a similar way. Neural networks are trained on enormous amounts of data, yet useful representations often become concentrated in relatively small latent structures. Much of modern deep learning can be interpreted as discovering compressed representations that preserve useful information.

If intelligence itself emerges through compression, then perhaps civilization has already been performing a version of this process for thousands of years.

The question then becomes:

If civilization has already compressed vast amounts of information, where are its most valuable representations located?

This is the motivation behind the earlier hierarchy:

Art → Literature → Social Science → Natural Science → Researchers → Engineers.

The hierarchy is not intended as a ranking of importance. It is a speculation about where information becomes increasingly structured and condensed.

Engineers are not less important than artists.

Rather, engineers typically operate on abstractions that have already passed through many previous layers of compression.

A successful engineering system may ultimately rest on scientific principles, which rest on conceptual frameworks, which rest on human observations, which themselves emerge from lived experience.

From this perspective, the central challenge for future LLM development may not simply be scaling data and compute.

It may be identifying which compressed representations of civilization contain the most useful information for intelligence.

This is what motivates the top-down perspective.

The question is not:

How do we replace engineering?

The question is:

Can civilization itself provide inductive biases that scaling alone cannot discover efficiently?



At first glance society appears completely disorganized.

Researchers move in one direction.

Entrepreneurs move in another.

Artists pursue different goals.

Engineers optimize different systems.

Politicians, scientists, writers, and investors all seem to operate according to unrelated incentives.

Yet when viewed from a sufficient distance, patterns emerge.

Civilization progresses despite the apparent chaos.

No single individual understands the entire system.

No central planner coordinates all activity.

Instead, progress emerges from a large number of independent attempts.

The implication for LLM research is interesting.

Perhaps no single discipline owns the answer.

Machine learning theory may hold part of it.

Systems engineering may hold part of it.

Neuroscience may hold part of it.

Philosophy may hold part of it.

Art may hold part of it.

The objective is therefore not to determine which discipline is "correct."

The objective is to understand how these perspectives interact.

Why Metrics May Eventually Fail

Current AI progress is largely measured through benchmarks.

Benchmarks are useful because they provide feedback.

Without feedback, optimization becomes impossible.

However, the most important properties of intelligence may not be benchmark-native.

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- ... wisdom,
- judgment,
 - taste,
 - creativity,
 - agency,
 - self-awareness.

None of these have universally accepted metrics.

Yet humans clearly recognize them when they encounter them.

This creates a paradox.

The qualities we care about most may be precisely the qualities that are hardest to measure.

Consequently, future progress may require evaluation frameworks that go beyond benchmark optimization.

This does not mean abandoning metrics.

It means recognizing that metrics are approximations rather than definitions.

The Long-Term Question

Ultimately, the most important question is not whether a model scores higher on AIME.

Nor is it whether a model writes better code.

Those are useful signals.

The deeper question is whether intelligence can be understood as a compressed representation of civilization itself.

If that is true, then the future of LLM development may depend not only on larger datasets and larger models, but also on understanding where civilization stores its most important information.

This is why discussions about art, philosophy, science, anthropology, machine learning theory, and engineering all belong in the same conversation.

They may appear unrelated.

But from a sufficiently high level, they are all attempts to answer the same question:

How does humanity approximate truth, and what role should intelligent systems play in that process?